Pulser amplitude	Amplifer gain	Amplifer amplitude	Channel number	Channel counts
Amp 1	gain 1			
Amp 2	gain 1			
...Amp 5	gain 1			
Amp 1	gain 2			
Amp 2	gain 2			
...Amp 5	gain 2			
Amp 1	gain 3			
Amp 2	gain 3			
...Amp 5	gain 3			

3.2.2 MCA Dead Time Measurements

- After every pulse which the MCA accepts and analyzes, it is 'dead' for a certain amount of time, i.e., it will not accept additional pulses during that time interval. The pulser can be used to measure the MCA dead time by using the double-pulse feature.

- PULSER: Set the rise time to be short
- PULSER: Set the fall time to be 10 µs.
- AMPLIFIER: Set the shaping time to 0.5 µs.
- Vary the pulse amplitude until the recorded pulses lie somewhere near the middle of the MCA channel range
- PULSER: Switch to double-pulse mode and observe the amplifier output on the scope while also supplying it to the MCA
- NOTE: The spacing between double-pulses can be varied using the 'delay' control on the pulser
- Start with the minimum delay on the pulser (or 10 µs).
- Slowly increase the delay until the MCA begins recording a distinct second peak corresponding to the smaller, delayed second pulse (in a channel far removed from the first one)
- NOTE: The double pulse feature of the pulser produces two pulses of the same amplitude. To produce pulses of differing amplitudes that can be resolved with the MCA, the pulser output must be shaped by the amplifier and then passed to the o-scope.
- Measure the corresponding time spacing on the scope which is now the corresponding MCA dead time. Repeat using several different initial pulse amplitudes and record the MCA dead time and the initial pulse channel number.

4 Required Plots

Create the following plots using your computer and the data you measured. Include them in your lab report.

Section 2.2.1

- Plot the SCA input amplitude vs. the corresponding LLD setting.
- Plot the SCA input amplitude vs. the rise time of the SCA input signal.

Section 2.2.2

- Plot the range of input pulse heights vs. the window width to check for linearity (plot all three LLD settings on the same graph).

Section 2.2.3

- Make necessary plots between appropriate variables based on your best judgement.

Section 3.2.1

- Plot the amplifer output amplitude vs. MCA channel number and net counts for three different amplifer gains.

Section 3.2.2

- Plot the MCA dead time vs. initial pulse channel number.

5 Post-Lab Questions

1. Why is integral linearity important?
2. How many SCAs would you need to equal the discriminatory capability of your specific MCA?
3. What are the advantages and disadvantages of increasing the number of channels?
4. Based on your results from the plot in section 3.2.2, what type of ADC is in your MCA?
5. Explain ballistic deficit and pulse pileup. Can you minimize the problems associated with both of these effects at the same time? Why or why not?